



SECP3843-01

TUTORIAL MS AZURE

LECTURER : DR ARYATI BINTI BAKRI

NO	NAME	MATRIC NO
1.	PRAVINRAJ A/L SIVABATHI	A23CS0171
2.	DHESHIEGHAN A/L SARAVANA MOORTHY	A23CS0072

1.0 Project Overview

1.1 Problem Background

In today's organizations, especially in the retail and sales organizations, millions of transactions are processed, about related products, and types of customers. But these data are usually in raw, unstructured form and are difficult to be directly analyzed to obtain useful knowledge. In the absence of good data pipeline, the organization might run into problems such as data inconsistencies, duplication, and slow processing of data, which may cause bad decisions and bad impact on overall business.

For this project, the sales data comprises information about the product categories, customer titles, and the sale value, all those need to be processed properly to get some insights. Traditional methods of processing data are not scalable and cannot effectively cope with the growth in amounts of data. Hence, there is a need to have a robust and automated data engineering solution to effectively handle, transform and analyze the data.

In addition, the traditional processing of data is not scalable with increasing data. This will require a powerful, automated data engineering solution that enables the management, transformation, and analysis of the data.

1.2 Project Objectives

The main objective of this project is to design and implement an end-to-end data engineering pipeline using Microsoft Azure. The specific objectives include:

- To automate the process of data ingestion from source systems
- To clean and transform raw sales data into structured and usable formats
- To implement a layered data architecture (Bronze, Silver, Gold) for better data organization
- To store and manage data efficiently using cloud-based storage
- To enable fast and efficient data analysis using analytical tools
- To create an interactive Power BI dashboard to visualize insights such as total sales, product distribution, and customer segmentation

1.3 Business Relevance

This work is particularly applicable to the retail, e-commerce and supply chain domains as they generate such transactional data in large volumes on a regular basis. Industries within these sectors need to know how they are selling, what their customers are doing, and how they can better run their businesses.

Using a cloud-based data pipeline, companies can store, run and analyze data faster. The project is about total sales at 708.69K, products, customer segmentation and so on through the dashboard. These insights enable improved strategic planning and decision making. This is consistent with current industry standards where data-driven decision making is necessary to remain competitive and to grow one's business.

1.4 Project Scope

The aim of this project is an end-to-end pipeline utilizing Microsoft Azure from data ingestion to visualization. The following are the key parts for this project:

- Data ingestion using Azure Data Factory
- Data storage in Azure Data Lake
- Data transformation using Azure Databricks
- Data analysis using Azure Synapse Analytics
- Data visualization using Power BI dashboard

The project specifically focuses on batch processing of structured sales data. The final output of the pipeline is an interactive dashboard that displays key metrics such as total sales, product count, and customer distribution. Advanced features such as real-time data streaming and machine learning integration are not included. This scope is defined to ensure a clear focus on core data engineering concepts while maintaining a manageable level of complexity.

2.0 The Solution

The solution developed in this paper is a full-stack data engineering pipeline, built on Microsoft Azure, for processing and analyzing sales data. The system converts raw sales data into actionable information, which can then be viewed via a dashboard containing various interactive visualizations. A Medallion Architecture of Bronze Silver and Gold is developed to guarantee the alignment of data within each layer. This architecture is also appropriate because the sales data have various fields including product categories, customer titles and sales values etc. and the data needs to be processed in a structured format before it can be analysed.

2.1 Data Ingestion

The process starts with ingestion using Azure Data Factory (ADF) to bring the raw sales data set into Azure Data Lake Storage. This is to make sure that everything, from product information to data related to customers, is in one place. ADF was selected as it automates the process of transferring data, saving the manual work and providing reliability in the processing of the dataset.

2.2 Data Storage

After ingestion, the sales data are kept in the Azure Data Lake serving as the consolidated storage. For this project, the data is layered for processing and analysis. This is crucial since the data has to be saved as raw and processed, so as to be able to do the right calculations for all the metrics so the total sales, product distribution and more is visible in the dashboard.

2.3 Data Processing

The data is transformed in a Medallion Architecture. The raw sales data are kept in the Bronze layer where no modification is performed, and it contains the original product categories and customer titles and so on. The Silver tier data is cleaned of duplicates and missing values the data is cleaned of missing and duplicate values. In the Gold tier, the data is aggregated and organized to produce meaningful measures like total sales (708.69K), number of products (397), and category-wise sales distribution. This tiered strategy guarantees the final data used for analysis is clean, reliable and well-tempered

2.4 Data Transformation

Transformation processing between the layers of the Medallion is done with Azure Databricks. It cleans and formats the sales dataset and creates the additional aggregates necessary for analysis. For instance, it aggregates sales information by product categories and customer titles to create the visualizations on the dashboard. Databricks is chosen due to its high performance processing of big data and flexible data transformation.

2.5 Data Analytics

With data Processing and structuring, the data is getting loaded to directly to azure synapse analytics for query and analysis. Synapse in this project permits for the rapid access to processed data used in creating dashboard insights. This is especially useful when we want to efficiently navigate through aggregated results such as category-wise sales , total sales . Synapse is preferred because it offers good analytics performance for structured business data.

2.6 Data Visualization

At the end, data is displayed via Power BI. The dashboard developed in this work offers important information, fulfilling above mentioned numbers for total sales 708.69K, while product count 397, category wise so on and customer title segmentation. We employ visual aid such as bar chart, pie chart to make the data easy to understand. This enables the user to sense sales performance trends and patterns more rapidly, which helps them make better decisions.

2.7 Overall Summary

There is a general pattern to the solution of first bringing in the data. I therefore had a solution that was logically flowing the whole way from data ingress via Azure Data Factory, storage in Azure Data Lake, processing via Azure Databricks utilizing the Medallion Architecture, querying via Azure Synapse Analytics, and ultimately visualizing with Power BI. Each element is rigidly integrated to ensure the production of genuine insights visualized in the dashboard. This organized flow allows for efficient processing of data throughout and mimics what would be implemented in data engineering scenarios in real world applications.

3.0 Technology use

3.1 Azure Data Factory

In this project, the sales dataset is injected into the data pipeline with Azure Data Factory. It fetches the raw data and loads it into Azure Data Lake Storage in batch mode. This matters for the project because the dataset has a number of features, like product category, customer title, sales values, that should be moved across consistently without the manual intervention in hand. ADF is selected as it provides dependable and repeatable data ingestion, which is key for maintaining the data integrity in the pipeline.

3.2 Azure Data Lake Storage

The sales data is stored at different stages of processing with Azure Data Lake Storage. It enables Medallion Architecture, where raw data resides in the Bronze layer, refined data in the Silver, and the aggregated sales in the Gold. This is very much related to this project as the dashboard can't run without structured, clean data to determine how much has been sold (708.69K) and how its products are distributed. The utilization of Data Lake is so that the sales data can be stored in high volume and processed flexibly.

3.3 Azure Databricks

The sales dataset is cleaned and manipulated by using Azure Databricks. In this project, it does data cleaning like dealing with null values, and duplicates and then some product and customer related data is structured. From another side, it also creates the categories based on which the sales values are analyzed. Databricks is used here as it allow for processing large amounts of data as well as scalable data transformations that allow for meaningful insights to be placed on the final dashboard.

3.4 Azure Synapse Analytics

After transformation, the sales data is stored and queried using Azure Synapse Analytics. In this project, we are facilitated with the queries option of structured data which will be used to visualize in the dashboard. This is important since the dashboard displays such aggregates as total sales, product count, and category breakdown, and these require fast query and optimized query performance. Synapse is selected because it is high performance analytic service that works well with structured business data.

3.5 Microsoft Power BI

Use Microsoft Power BI to construct the final dashboard data visualization. In this project, it provides overarching summary information such as total sales (708.69K), number of products (397), category composition and customer title distribution. The dashboard you create can contain visual objects such as bar and pie charts to facilitate the interpretation of the data. Power BI is selected as it provides interactive visualisation that helps users to drill down the data and fetch out trends and patterns.

3.6 Integration of Azure Services

In this project, you'll work with multiple Azure services to build an end-to-end data pipeline that ingests raw sales data and outputs actionable insights. The flow is: Ingest data with Azure Data Factory, store it in Azure Data Lake. Data from Azure Databricks is processed and refined, then stored and queried in Azure Synapse Analytics. Lastly, the processed data is visualized by Power BI as an interactive dashboard. This integration provides seamless, stream data flow to deliver accurate business insights led on the dashboard directly.

4.0 Data Pipeline Architecture

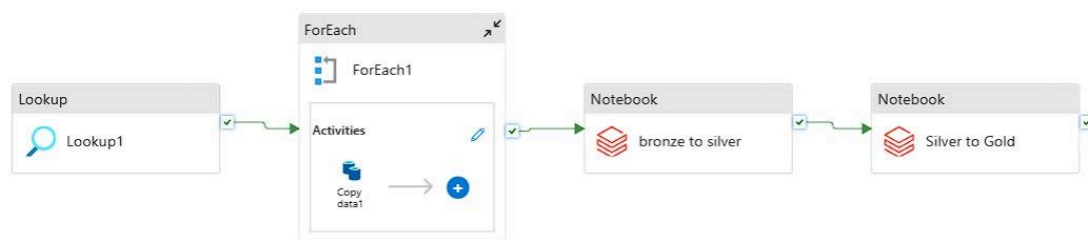


Figure 1 shows Data Pipeline Architecture of Sales Data

The data pipeline of this project is developed to flow sales data among the processes of data ingestion, storage, transformation, analytics, and visualizations. The processing architecture has a well-organized and layered structure to allow smooth transformation of raw data into useful information. It combines multiple Microsoft Azure services, into an end-to-end workflow to enable a dashboard visualization of sales metrics: total sales (708.69K), product count (397), and category distribution.

4.1 Data Source

The source data for the pipeline is a sales dataset that includes product categories, customer titles, and sales values. This unprocessed dataset is the input to the whole pipeline. Significance of the Data Source The data source is significant as it contains all the primary information needed to extract meaningful insights using the dashboard.

4.2 Data Ingestion Layer

In the process of ingestion, Azure Data Factory is leveraged to extract the raw sales data from the source and dump it in Azure Data Lake Storage. This layer guarantees that data are gathered uniformly and automatically. ADF has major contribution to data flow stability and provides support for scheduling so that the pipeline can be run multiple times without any intervention.

4.3 Data Storage Layer

After ingestion, the data is stored in Azure Data Lake in Medallion Architecture. In the Bronze level, the raw data is stored as it is, with the original records. In the Silver layer, the data is cleaned and transformed to fix errors, remove duplicates, and inconsistencies. At the Gold level, data is summarized and formatted in a way that can be readily analyzed. This modelled data is then used for the visualization or reporting layer in our case dashboard report.

4.4 Data Transformation Layer

The transformation layer is Azure Databricks whereas the data processing and refining during the three Bronze, Silver & Gold stages is out of scope of this tutorial. For this project, you will be using Databricks to clean the data set, to organize product and customer information, and to compute aggregated values such as total sales and sales by category. This process is critical to make sure that the data you use for analysis are correct and meaningful.

4.5 Data Analytics Layer

After transforming the data and saving it in the Gold layer, now load it into Azure Synapse Analytics. Querying and analyzing structured data from layers. Synapse also supports the rapid querying of this aggregated data, so that insights generated by the dashboard are displayed in Power BI. It provides you with robust performance analytics when you are processing a huge number of sales data.

4.6 Data Visualization Layer

The Layered Architecture The final component to the architecture is the presentation layer with Power BI that allows you to publish a rich and interactive report. The dashboard offers at a glance key information like gross sale (708.69K), product count (397), category share, customer title share. Visuals such as pie charts and line graphs help you understand the data and see patterns. Here processed data is converted into actionable insights for decision making layer.

4.7 End-To-End Data Flow Explanation

In general, the data goes through pipeline sequentially from the data source, to Azure Data Factory for ingestion, to Azure Data Lake for storing at multiple layers(bronze, silver, gold). The data is processed and transformed in Azure Databricks and analyzed in Azure Synapse Analytics. And then Power BI to display the data as an interactive report and dashboard. This end to end design facilitates a seamless and high perform the data flow for the generation of reliable, real-time insights.

5.0 MS Power BI Dashboard

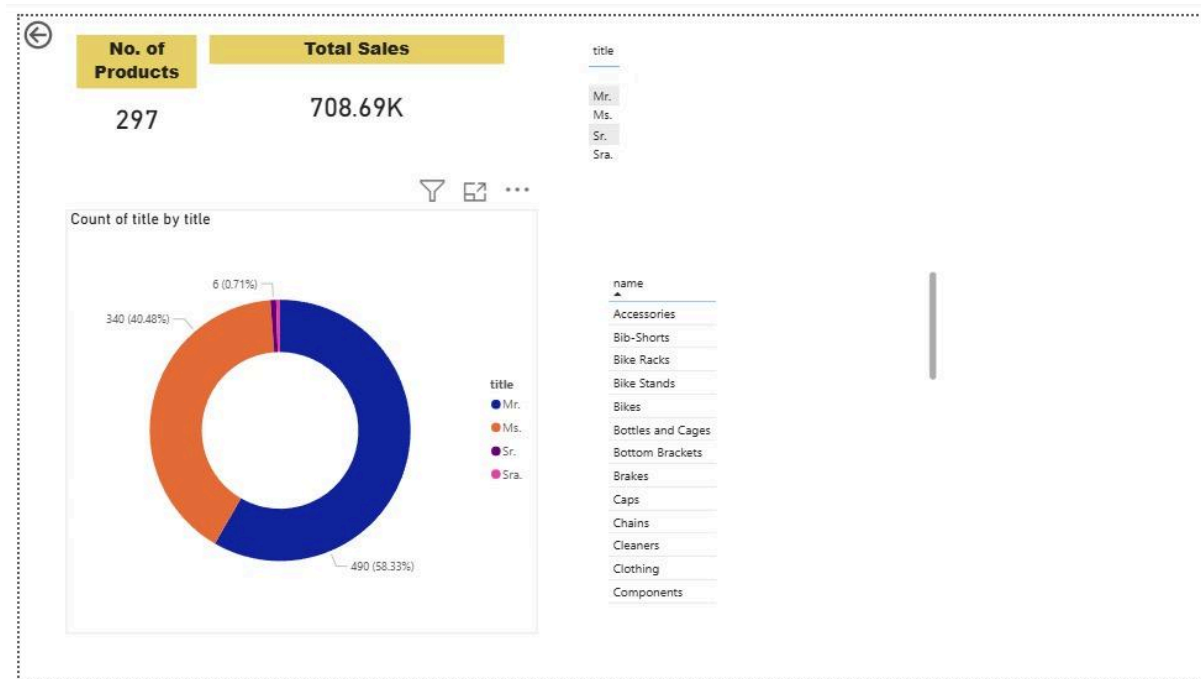


Figure 2 shows dashboard of the sales data

The proposed Microsoft Power BI sales analysis dashboard provides processed sales data in an attractive and interactive manner. It is intended to deliver high level information about business that is easy to scan and enables user to have a good understanding of it at high level. The total sales, number of products, customer clusters, and product type are all key features of the dashboard, and it is relied upon for data-based decisions.

5.1 KPI Cards (Key Performance Indicators)

KPI cards at the top of the dashboard display key summary numbers like Total Sales (708.69K) and Number of Products (297). These cards allow users to gain a high-level understanding of business performance without having to review complex charts. To quickly get an insight on the overall sales and product volume, these cards are used as KPI cards for the sales and product volume.

5.2 Donut Chart

A donut chart is used to show the distribution of customer titles such as Mr, Ms, Sr and Sra. This is chart to visualize the percentage of each customer group in the sample. As an illustration, the chart indicates that most of the customers are “Mr.”, then followed by “Ms.”, and all the other less percentages of these bought candies. This chart is designed to give you an understanding of what your customer is like, which can be useful in customer marketing and segmentation.

5.3 Category/Product

On the right of the dashboard, there is a category or product list (slicer) which contains such items as Accessories, Bikes, Clothing and Components. This enables you to filter the dashboard by product categories. The main purpose of the slicer is to provide the dashboard users with greater interactivity, allowing them to dive into the data, and to provide a more flexible/dashboard experience.

5.4 Data Storytelling and Insights

Together, the KPI cards, donut chart, and slicer tell a simple data story. Users can then drill down into the KPI cards, donut chart and slicer to analyze the customers’ distribution and favorite product categories. Say a user wants to slice by a sub-category like “Bikes” to see how that segment contributes to total sales and customers. This makes it easier to tell which categories do better and which customer segments contribute the most to your sales.

5.5 Effectiveness of Visualization Design Quality

The visualizations on the dashboard are friendly to eyes. KPI cards provide quick summaries, donut chart shows proportional data well, and slicer add some interactivity. The design is simple and sleek, which makes it easy for users to analyze the information. All these elements have their own specialized purpose that makes the dashboard informative yet simple to use.

5.6 Summary of Dashboard Contribution

The result is that the Power BI dashboard effectively translates processed data into valuable actionable intelligence. With a mix of KPI indicators, charts, and interactive slicers, users can drill down on sales, customer distribution, and product categories. This illustrates the output of the data pipeline and shows how visualization of data can be used to drive business decisions.

6.0 Reflection

Reflection by Pravinraj

Throughout this project, I gained valuable experience in building an end-to-end data engineering pipeline using Microsoft Azure. I was mainly involved in understanding and implementing the data ingestion and storage process using Azure Data Factory and Azure Data Lake. Initially, I found it challenging to understand how different Azure services connect and work together, especially when configuring the pipeline and ensuring the data flows correctly between each stage. However, by following the tutorial and experimenting with the tools, I was able to gradually understand the workflow.

Among the biggest challenges I had to deal with was how to process errors in data ingestion and full proper loading of the dataset into the storage layer. I got some ideas how to fix these things by looking at configurations and using a little of command line pipe line testing. That really helped me to know how to solve the problems and gave me more confidence in cloud systems.

For this project I got technical experience with ADF and ADL and for data pipeline in general. I also learned the importance of pre-structured/processed data before analysing it. Overall this experience gave me a good insight into how raw data is handled and pre-processed in real life data engineering problems.

Reflection by Dheshieghan

This project gave me a deeper understanding of how data engineering concepts are applied in a practical environment. Throughout the project, I worked with tools such as Azure Data Factory, SQL Server, Databricks, and Power BI, while learning how data is extracted, transformed, stored, and visualized through a complete data pipeline.

During the project, I faced several challenges, especially in configuring the Self-hosted Integration Runtime, solving SQL connection issues, and troubleshooting Java Runtime errors for Parquet processing. Although these problems took time to resolve, they helped improve my problem-solving skills and taught me the importance of patience and troubleshooting in real-world projects.

I also learned how the Medallion architecture works through the Bronze, Silver, and Gold layers, and how each stage helps improve data organization and readiness for analysis. In addition, building the Power BI dashboard helped me understand how data can be presented in a more meaningful and interactive way.

Overall, this project was a valuable learning experience that strengthened both my technical knowledge and confidence in working with data engineering tools. It also showed me the importance of continuous learning and adaptability when working with complex systems.

